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EDITORS

Assist. Prof. Dr. Ümit KARAKAŞ
Assist. Prof. Dr. Nagihan KÖROĞLU KABA

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METHODS OF RECORDING OCCLUSION UNCTIONAL OCCLUSAL AND VIRTUAL ARTICULATORS IN PROSTHODONTIC

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ABSTRACT
This seminar discusses the evolution and methodologies in recording occlusion and utilizing virtual articulators in prosthodontics. Traditional methods like wax bite registration and articulating paper, although essential, have limitations in capturing dynamic occlusion and occlusal forces. With digital dentistry advancements, the introduction of intraoral scanners and virtual articulators has revolutionized the precision of dental restorations. This seminar outlines the methodologies, tools, and clinical applications of both conventional and modern approaches, emphasizing accuracy and patient comfort
Keywords: occlusion, virtual articulators, wax bite registration, intraoral scanners, digital dentistry, occlusal analysis
Background: Occlusion refers to the static and dynamic relationship between the upper and lower teeth during functional movements such as biting and chewing. Proper occlusal relationships prevent issues such as temporomandibular joint (TMJ) disorders and ensure the even distribution of occlusal forces. Traditionally, occlusal recordings were performed using wax bite registrations and articulating paper to assess contact points. However, the static nature of these techniques limits their ability to capture real-time dynamic occlusion, leading to a need for advanced methodologies .
Purpose: The purpose of this seminar is to explore both conventional and digital techniques for recording occlusion in prosthodontics. Emphasis is placed on comparing the accuracy and clinical outcomes between traditional methods and modern technologies like virtual articulators and intraoral scanners. This seminar aims to guide prosthodontists in integrating these techniques to enhance diagnostic precision and treatment outcomes .

INTRODUCTION
Occlusion is morpho-physiologically static, dynamic, and uniaxial and can be altered by modifications in the stomatognathic system, teeth, muscles, ligaments, and temporomandibular joint, it is also defined by the Glossary of Prosthodontics as 'the static relationship between the incising or masticating surfaces of the maxillary or mandibular teeth or tooth analogues (Azpiazu-Flores and Mata-Mata, 2021). An inadequate understanding and poor management of occlusion can result in various complications, including pain/discomfort, tooth mobility, tooth wear, fractured teeth or restorations, and TMJ dysfunction (Velásquez *et al.*, 2023). Many of these complications can be mitigated, or avoided, when the occlusion is carefully controlled, dentists have employed various methods to record occlusal relationships, each offering unique advantages and insights into occlusal dynamics. Traditional techniques, such as wax bite registration, articulating paper, and foil registration, have stood the test of time, providing tactile feedback and visual assessment of occlusal contacts, However, traditional methods may have limitations (Ferencz, J. L., & Silva, N. R. (2015).

With the advent of digital dentistry, new avenues have emerged for occlusal recording and analysis, intraoral scanners, 3D scanners, and computer-aided design/computer-aided manufacturing (CAD/CAM) systems enable precise digital impressions and virtual occlusal analysis, revolutionizing the way clinicians evaluate occlusion (Lassila, L. V., & Tezvergil-Mutluay, A. 2020).

Occlusion analysis methods are used to assess how teeth come into contact during biting, chewing, and other jaw movements. The goal is to identify any misalignments or issues in how the upper and lower teeth interact, which can affect oral function, comfort, and long-term health of the teeth and temporomandibular joint (TMJ), key methods include, articulating paper, bite registration (wax/silicone), T-Scan, EMG, CBCT, bite Force Sensors, virtual articulators.

A virtual articulator is a digital tool that simulates a physical articulator, allowing dentists to analyze and visualize a patient's jaw movements and occlusion in a virtual environment. It uses 3D scans, often combined with CBCT and intraoral data, to create a precise digital model of the patient's bite (Ghodasra and Brizuela, 2023).

METHOD

A comprehensive review was conducted on various occlusion recording techniques. Traditional methods, including wax bite registration, articulating paper, and foils, were compared against digital techniques such as intraoral scanners, digital jaw tracking systems, and virtual articulators (Madanshetty *et al.*, 2023).

The study also evaluated the performance of different recording materials like polyvinyl siloxanes (PVS) and computerized occlusal analysis systems like T-Scan (Camcı and Salmanpour, 2021).

Functional occlusal analysis methods such as electromyography (EMG) and joint vibration analysis were explored for their roles in assessing occlusal harmony (Revilla-León et al., 2023).

RESULTS

Digital occlusion recording methods demonstrated superior accuracy and reduced operator error compared to traditional techniques. Intraoral scanners allowed for precise digital impressions, while virtual articulators provided detailed simulations of mandibular movements, enhancing prosthetic designs. The use of polyvinyl siloxane materials improved the dimensional stability of occlusal records. Furthermore, computerized occlusal analysis systems like T-Scan proved effective in capturing real-time occlusal dynamics and occlusal forces (Yazigi *et al.*, 2023).

Group	Product	Material	Manufacturer	Lot.Nr.
RC	Registrado X-tra	Polyvinyl siloxane	Voco	1742109
RS	Registrado Scan	Polyvinyl siloxane, scannable	Voco	1735242
FC	Futar D Fast	Polyvinyl siloxane	Kettenbach	171131
FS	Futar Cut & Trim Scan	Polyvinyl siloxane, scannable	Kettenbach	160021
OC	O-Bite	Polyvinyl siloxane	DMG	778917
OS	O-Bite Scan	Polyvinyl siloxane, scannable	DMG	768094

Table 1. Six groups of the interocclusal recording materials, 3 conventional interocclusal registration materials (RC, Registrado X-tra [Voco]; FC, Futar D Fast [Kettenbach]; and OC, O-Bite [DMG]), 3 scannable interocclusal registration materials (RS, Registrado Scan [Voco]; FS, Futar Cut & Trim Fast [Kettenbach]; and OS, O-Bite Scan [DMG]), (Yazigi *et al.*, 2023).

Manufacturer	Occlusal analysis system	Functionality	Occlusal sensor	Occlusal sensor thickness
Bausch	OccluSense	Static and dynamic occlusal contacts (pressure and timing)	OccluSense Electronic Pressure Sensor (Large and extra-large sizes)	60 µm
Dmtec	Accura	Static and dynamic occlusal contacts (pressure and timing)	Occlusal sensor (S and M sizes)	160 µm
Tekscan	T-scan Novus	Static occlusal contacts (pressure and timing) Total bite force (N)	T-scan sensor (S and L sizes)	100 µm

Table 2. Description of the main commercially available computerized occlusal analysis systems (Revilla-León et al., 2023)

DISCUSSION

The evaluation and recording of occlusion have undergone significant transformation from traditional analog methods to contemporary digital technologies. This evolution has been driven by the need for precision, efficiency, and enhanced patient care.

Traditional methods, such as wax bite registrations and articulating papers, have long served as foundational techniques in prosthodontics. While these methods are cost-effective and straightforward, their limitations in capturing dynamic occlusal relationships and force distributions have prompted the integration of advanced tools and materials (Sibel Ezgi Bostancıoğlu, Abdullah Toğay and Evşen Tamam, 2021).

Recent advancements in elastomeric bite registration materials, including polyvinyl siloxane and polyether, have addressed some of these limitations by offering improved accuracy and dimensional stability. These materials have proven invaluable in fixed and removable prosthodontics, enabling clinicians to achieve more predictable outcomes. However, even these advancements are challenged by the growing demands for precision in complex restorative and orthodontic cases (Yazigi et al., 2023).

Digital dentistry, spearheaded by intraoral scanners (IOS), virtual articulators, and computerized occlusal analysis systems like the T-Scan, represents a paradigm shift in occlusal recording. These technologies allow for a more comprehensive analysis of both static and dynamic occlusal relationships, facilitating accurate treatment planning and the creation of restorations. IOS systems, for instance, integrate maxillomandibular relationships into virtual diagnostic casts with remarkable precision, although operator skills and patient factors remain critical determinants of success (Kihara *et al.*, 2020).

Furthermore, computerized occlusal analysis systems provide quantifiable insights into occlusal forces and timing, which are unattainable with conventional methods. Despite their potential, limitations such as sensor durability and cost present barriers to widespread adoption.

The introduction of functional occlusal analysis methods, including masticatory muscle activity and joint vibration analysis (JVA), adds another layer of diagnostic capability. These techniques not only provide insights into occlusal harmony but also highlight potential temporomandibular joint (TMJ) dysfunctions and masticatory inefficiencies (Vasavi et al., 2023).

Incorporating virtual articulators into prosthodontics exemplifies how digital tools can optimize treatment workflows. These devices simulate jaw movements and occlusal relationships with high fidelity, enabling clinicians to anticipate and address potential issues before fabrication. However, the lack of universally accepted protocols and the steep learning curve associated with digital tools underline the importance of further research and training (Doshi et al., 2024).

In summary, while traditional methods remain relevant, the integration of digital tools enhances diagnostic accuracy, treatment planning, and patient outcomes. The convergence of technology and clinical expertise is essential for harnessing the full potential of these advancements in occlusal recording and analysis.

CONCLUSION

The advancements in recording occlusion and utilizing virtual articulators in prosthodontics have revolutionized the precision and efficiency of dental restorations. Traditional methods of occlusion recording, such as wax bites, facebows, and mechanical articulators, have been fundamental in providing accurate interocclusal records. However, these methods are often time-consuming and prone to human error. The introduction of digital technologies has addressed many of these challenges (Sibel Ezgi Bostancıoğlu, Abdullah Toğay and Evşen Tamam, 2021).

Digital occlusion recording methods, including intraoral scanners and digital bite registration systems, offer enhanced accuracy and convenience. These tools capture precise occlusal details and allow for immediate analysis and adjustments.

Furthermore, the integration of digital records with virtual articulators has streamlined the workflow in prosthodontics (Emanuele Risciotti et al., 2024).
Virtual articulators replicate the movements of the temporomandibular joint (TMJ) and the dynamic relationship between the maxillary and mandibular arches. They enable prosthodontists to simulate various occlusal schemes and functional movements with high fidelity. This simulation capability facilitates the design and fabrication of dental prostheses that are aesthetically pleasing and functionally robust (Doshi et al., 2024).
Moreover, the combination of digital occlusion recording and virtual articulation has significantly enhanced the diagnostic and treatment planning process. It allows for better visualization, more accurate predictions of treatment outcomes, and improved communication with patients and dental technicians (Özdemir *et al.*, 2021).

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